

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

MARKING SCHEME

NOVEMBER 2015

CHEMISTRY

9189/1

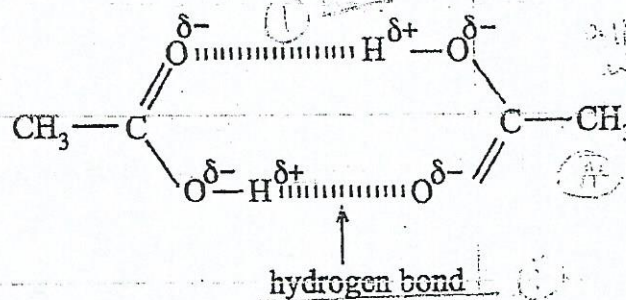
- 1 (a) B.P increases from PH_3 to SbH_3 because Van der Waals forces become stronger with increase in molecular size/ electron cloud/ AW; [1]

NH_3 has an anomalously high b.p. because of stronger hydrogen bonds between NH_3 molecules; [1]

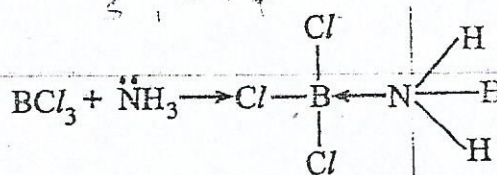
- (b) (i) Methanol can form hydrogen bonds with water molecules/ [1]

Octanol has a large hydrophobic (octyl) group; [1]

- (ii) Acetic acid dimerises; due to hydrogen bonding; [2]



- (iii) Electron deficient boron atom in BCl_3 accepts the lone pair on the nitrogen atom of NH_3 to form a dative bond.



C atom in CCl_4 has no vacant orbital to accommodate the lone pair of electrons from NH_3 ; [2]

- (iv) Lone-pair to bond-pair repulsion between 1 lone pair and 3 bond pairs in PCl_5 ; [1]

is greater than bond pair to bond pair repulsion between the 3 bond pairs in BCl_3 ; [1]

BCl_3 has 3 bond pairs and 0 lone pairs whilst PCl_5 has 3 bond pairs and 1 lone pair;

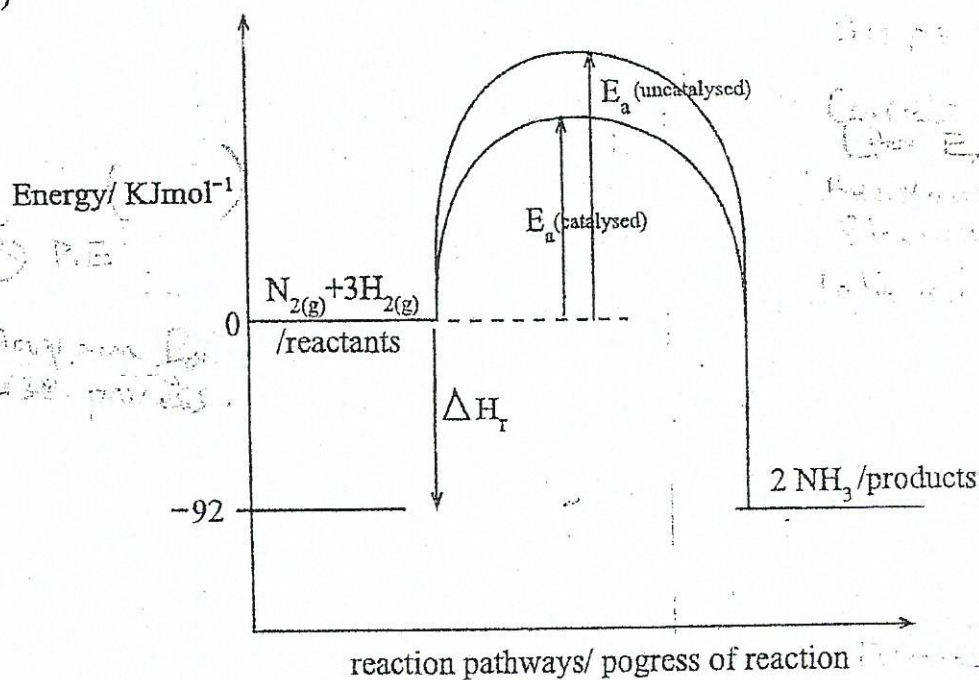
- (v) CO_2 is giant molecular / ionic with strong covalent / ionic bonds; [2]

[Total: 12]

- 2 (a) (i) 1. Activation energy is the minimum amount of energy required for a chemical reaction to occur/AW: [1]
2. A catalyst is a substance which alters the rate of a reaction and remains unchanged at the end of the reaction/ AW: [2]

- (ii) A lot of energy (994 kJ mol^{-1}) is required to break the strong $\text{N} \equiv \text{N}$ bond;

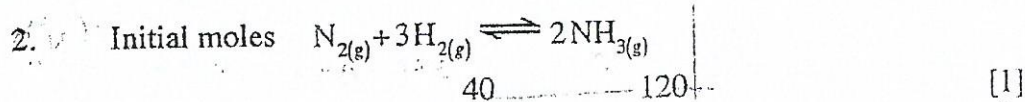
(b) (i)



(ii) E_a uncatalysed reaction = $+335 - 92$

= $+243 \text{ KJ mol}^{-1}$; [1]

(c) 1. $K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$; [1]



Equilibrium moles $40 - x$ $120 - 3x$ $2x$;

= $40 - \left(\frac{40}{100} \times 40\right)$ $120 - \left(\frac{40}{100} \times 120\right)$ 32 ; [1]

= 24 = 72

Concentration

$$/\text{mol dm}^{-3} \quad \frac{24}{2} = 12; \quad \frac{72}{2} = 36; \quad \frac{32}{2} = 16;$$

[1]

(NB $x = 16$)

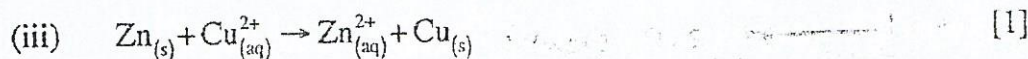
$$K_c = \frac{16^2}{12 \times 36^3} = 4.57 \times 10^{-4} \text{ mol}^{-2} \text{ dm}^6;$$

[2]

[Total:12]

- 3 (a) (i) The potential difference set up between two connected half cells at 1 atm, 298 K and 1 M concentrations of aqueous solutions. [1]

$$\begin{aligned} \text{(ii)} \quad E^\circ_{\text{cell}} &= E_{\text{red}} - E_{\text{oxid}} / E_{\text{RHS}} - E_{\text{LHS}} \\ &= +0.34 - (-0.76) \\ &= +1.10 \text{ (volts)} \end{aligned} \quad [1]$$

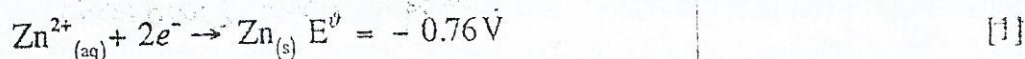


- (iv) E° increases since the equilibrium of the zinc half cell shifts to the right.



- (b) Zinc impurities dissolves from impure anodes since E° for the reaction $\text{Zn}_{(s)} \rightarrow \text{Zn}^{2+} + 2e^-$ is positive, $E^\circ = +0.76 \text{ V}$ [1]

The Zinc ions remain in solution since E° for their reduction is -ve.



- (c) (i) $\text{Zn} + 2\text{Fe}^{3+} \rightarrow \text{Zn}^{2+} + 2\text{Fe}^{2+}$ [1]
 $5\text{Fe}^{3+} + \text{MnO}_4^- + 8\text{H}^+ \rightarrow 5\text{Fe}^{2+} + \text{Mn}^{2+} + 4\text{H}_2\text{O}$

$$\text{(ii)} \quad \text{Moles MnO}_4^- = 0.0200 \times \frac{100}{1000}$$

$$= 0.00200 \quad [1]$$

$$\text{Moles Zn} = \frac{5}{2} \times 0.0020$$

$$= 0.00500 \quad [1]$$

$$\begin{aligned}\text{Mass of Zinc} &= 0.00500 \times 65.4 \\ &= 0.327\end{aligned}$$

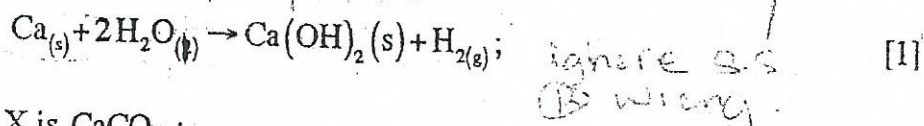
$$\begin{aligned}\% \text{ purity} &= \frac{0.327}{0.350} \times 100 \\ &= 93.4\%\end{aligned}$$

[1]

[Total: 12]

4 (a) (i) Calcium burns with a ^(vigorously) bright / brilliant ^(crimson / brick red) flame forming a white residue; ^{Both observations} $\text{Ca(s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CaO(s)}$ [1] + [1]

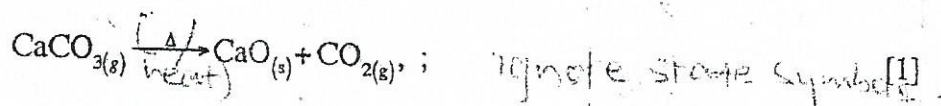
(ii) Ca reacts readily with cold water giving a steady evolution of hydrogen gas bubbles and a white suspension of Ca(OH)_2 ; ^{white precipitate} [1]



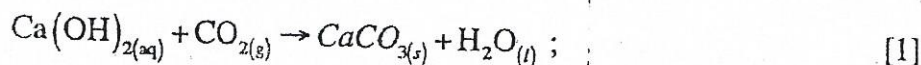
(b) (i) X is CaCO_3 ; [1]

(ii) A hissing sound/ steam is produced as the water touches the lime/ the lime swells/ crumbles to a powder with a larger volume; [1]

(iii) Reaction A



Reaction D



(iv) ^{Mg(OH)_2 less soluble than Ca(OH)_2 / AW - only thing to do is solubility} White suspension with Ca(OH)_2 whereas thicker white precipitate/ solid with Mg(OH)_2 ; [1]

(c) Calcium carbonate decomposes at a higher temperature/ AW: [1]

^{smaller} Smaller Mg^{2+} has a lower charge density than the ^{larger} Ca^{2+} ion; / AW [1]

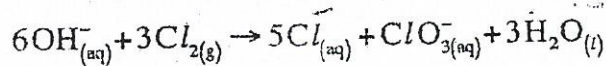
^{larger} Larger CO_3^{2-} less polarized by Ca^{2+} (to form smaller O^{2-} ion) / AW; [1]

[Total: 12]

5 (a) (i) NaOH is used in making of soaps/ detergents/ bleaches/ degreasing fluids/ weed killers; *any one use* [1]

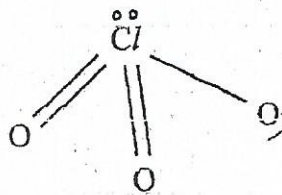
(ii) To avoid backflow of NaOH_(aq) to the anode compartment where it would react with the chlorine (in a disproportionation reaction)/AW; *use cathode for making of sodium hydroxide* [1]

(b) (i) $6\text{NaOH}_{(aq)} + 3\text{Cl}_{2(g)} \rightarrow 5\text{NaCl}_{(aq)} + 3\text{H}_2\text{O} + \text{NaClO}_{3(aq)}$ / [1]

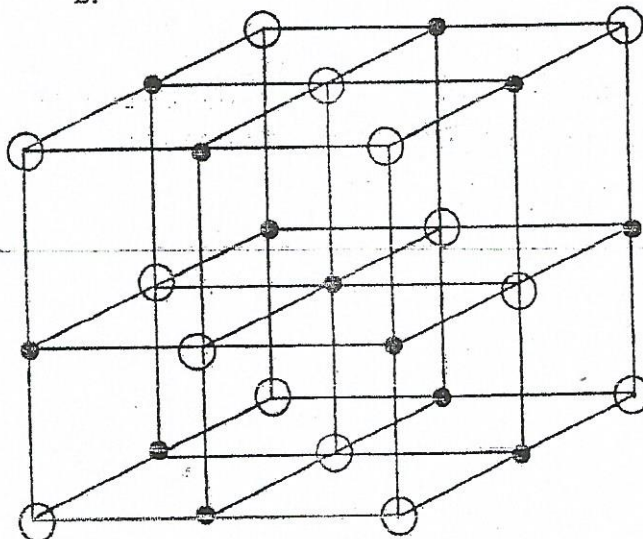


(ii) disproportionation; *disproportionation* [1]

(iii) F



2.

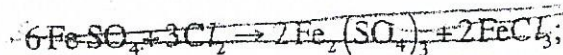
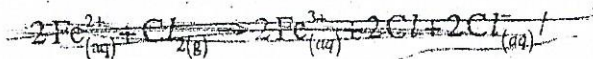


● Na⁺ ion

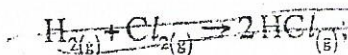
○ Cl⁻ ion

more 3D unit cell
more grey

(c) (i) Solution of iron (II) sulphate turns red - brown (from pale green) [1] *accept yellow/orange/side*
③ Cl₂ colour change



(ii) Chlorine burns with a pop/ Cl₂ and H₂ explode in sunlight or uv light; [1]



[1]